

School Club Manager

Challenge 1

Store club records and search them with functions.

Build it

1. Create a 2D `list` called `clubs` with rows `["art", 12, "R1"]`, `["chess", 18, "R2"]` and `["code", 15, "R3"]`.
2. Write `show_records(records)` to display each club's name, members and room on one line.
3. Write `find_members(records, target)` so it will `return` the matching club's member count, or `return -1` after the loop when nothing matches.
4. In the main program, call `show_records()`, ask for a club name, use `lower()` on it and call `find_members()`. Display `"Members:"` followed by the result, or `"Club not found."` for -1.

Test it

- `"Art"` displays `"Members: 12"` and `"code"` displays `"Members: 15"`.
- `"music"` displays `"Club not found."`

Talk about it

Why must `return -1` come after the loop and not inside it?

Save as `challenge1.py`.

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Challenge 2

Add a new club and change a member count.

Build it

1. Write `add_record(records)` to ask for a club name, a whole-number member count and a room. Use `lower()` on the name, then append the new row to records.
2. Write `update_members(records, target, new_members)` to change the matching club's member count and `return True`, or `return False` after the loop when nothing matches.
3. In the main program, call `add_record()`, then ask which club to update and its new count. Display "Club updated." or "Club not found." from the returned value, then call `show_records()`.

Test it

- Add "Music", 9, "R4", then update "chess" to 20. Both changes appear.
- Updating "drama" displays "Club not found." and no record changes.

Talk about it

`add_record()` does not `return` anything. How does the new row still reach `clubs`?

Save as `challenge2.py`.

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Challenge 3

Put all the functions behind a repeating menu.

Build it

1. Store `True` in `running`. Use a `while` loop to display "1: show, 2: search, 3: add, 4: update, 5: exit" and ask for a choice.
2. For choices 1 to 4, ask for any extra input and call the matching function.
3. For "5", store `False` in `running`. For anything else, display "Choose a valid option."
4. After the loop, use `len()` to display "Final record count:" followed by the number of rows.

Test it

- 3 to add "music", 9, "R4", then 4 to update "music" to 11, then 1 and 5: the music row shows 11 and "Final record count: 4" appears.
- "x" then 5: "Choose a valid option." then "Final record count: 3".

Talk about it

Why does every function take `records` as a parameter?

Save as `challenge3.py`.